

COMP 3023 – Design Patterns with C++ (SP2 2023)

Course	Design Patterns with C++
Name	Gitae Bae
Email	baegy002@mymail.unisa.edu.au
Identifier	110310861
Date	Sunday, 11/06/ 2023

Assignment2 – Programming Assignment 2: Patient Management System Partial Implementation

[Gitae Bae]

[11/06/2023]

Contents

Design Structure with UML diagram	3
Functional requirement FR1: Load patients from file.....	4
Functional requirement FR2: calculate the patient alert levels.....	5
Functional requirement FR3: calculate the alert level for all diseases a patient has.....	6
Functional requirement FR4: alert the hospitals and GPs	8

Design Structure with UML diagram

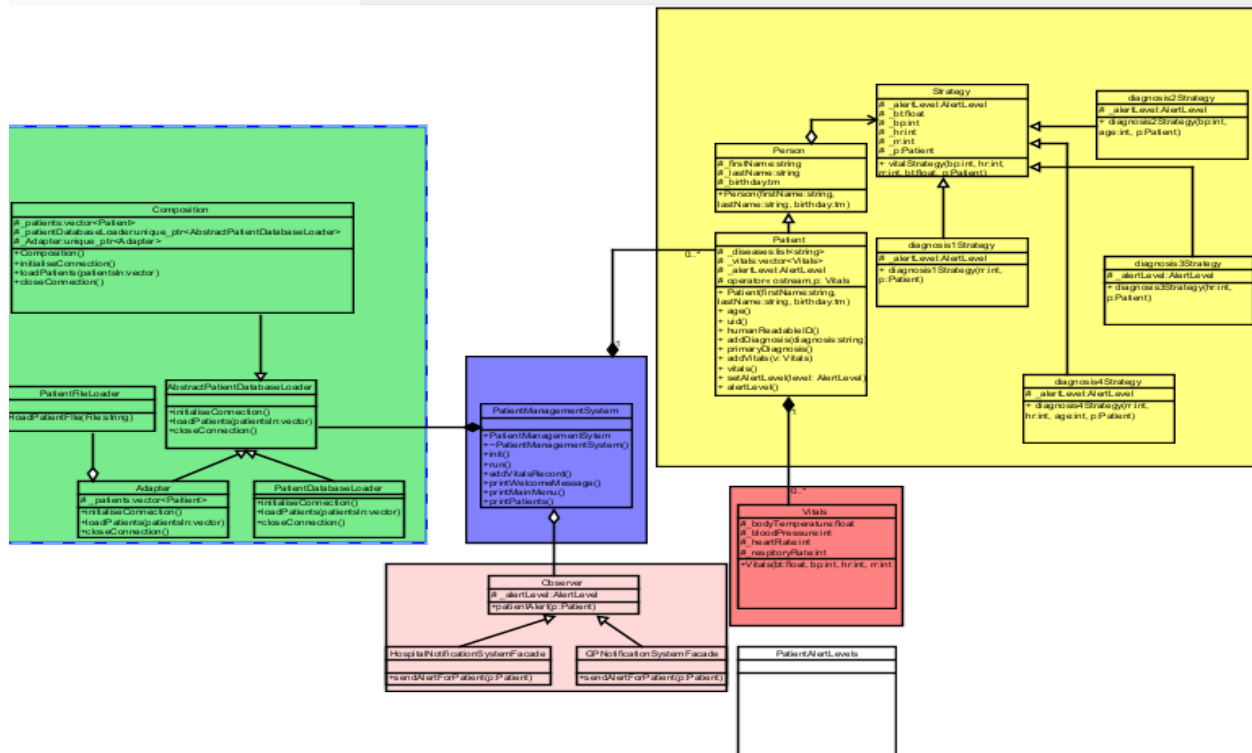


Figure1: Whole Design Structure

UMLdiagram -
design.uvx

Functional requirement FR1: Load patients from file

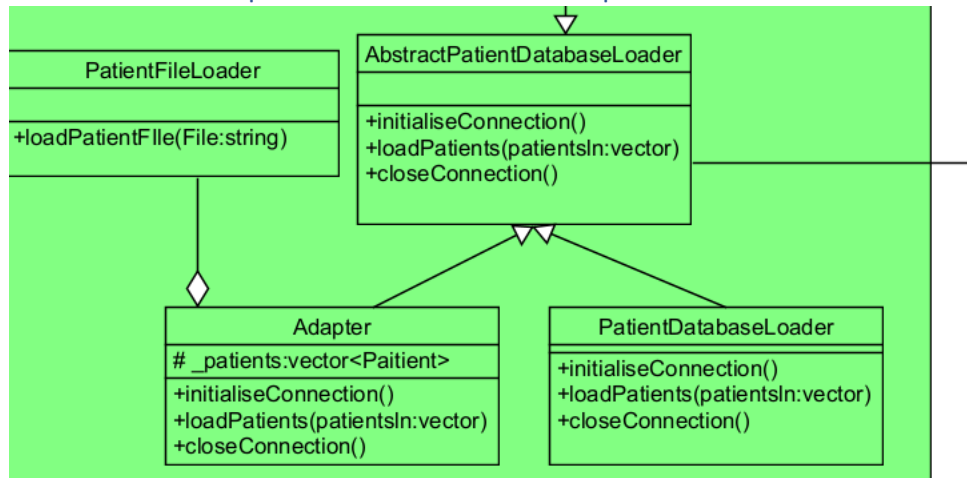


Figure 2: Adapter Pattern

Design Patterns: Adapter Pattern

PatientFileLoader class to load patient data from the file using Adapter design pattern. So, consider what design pattern is able to be used to address this problem and get the design class diagram to factor in the pattern and how to complete the design changes to address the problem. I used Adapter class using AbstractPatientDatabaseLoader to change the interface of PatientFileLoader.

How it works:

1. Connect initialiseConnection(), closeConnection(), patientLoad().
2. Patient.txt file connected to Adapter.cpp file.
3. Patient file load can be executed in PatientFileLoader.
4. PatientFileLoader can read the patient detail information and save it in vector.

Git commits:

1. I first added Adapter class in commit 612763.
2. I fixed something in commit 8a0e00b.

```

commit 61276369649ea2435b33e957e12233e9b51d7fbc
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date: Mon Jun 5 20:02:25 2023 +0900

    F1~F4- Draft of Design Pattern.

commit 8a0e00b43548b3882680611879ed2ef843a76662
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date: Thu Jun 8 20:13:30 2023 +0900

    F1 - Adapter and Composite design pattern
  
```

Functional requirement FR2: calculate the patient alert levels

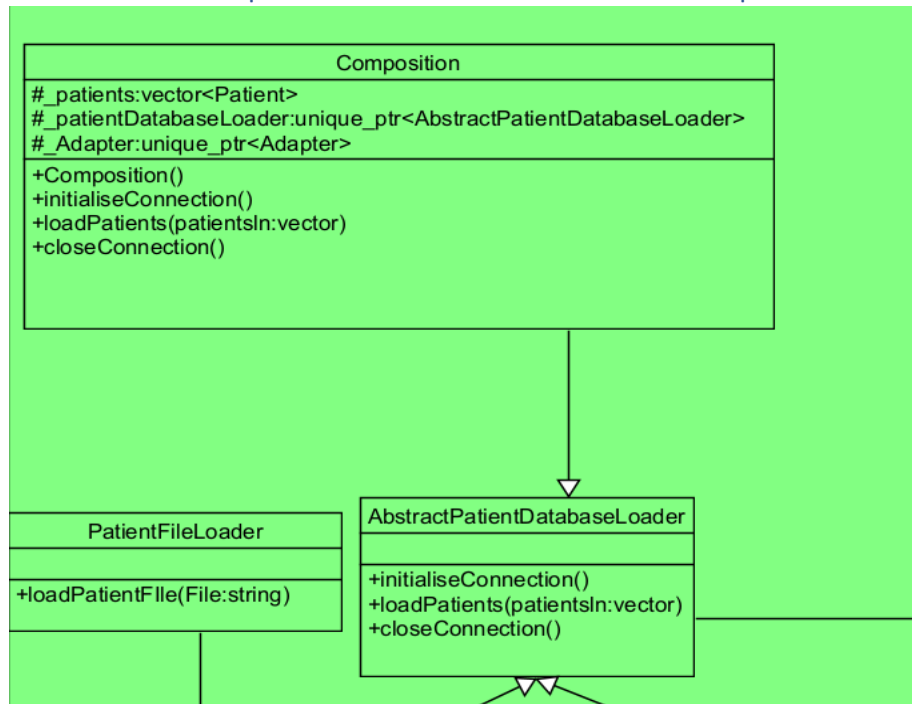


Figure3: Composite Pattern

Design Patterns: Composite Pattern

The system has to use the correct algorithm for calculating a patient's alert level dependent on the patient's primary disease. Each patient uses an algorithm dependent, and the alert level should be calculated using Vitals and not be calculated for patient data loaded and consider what design pattern and complete the design changes. Composite file can load AbstractPatientDatabaseLoader to process data and file together.

How it works:

1. Connect initialiseConnection(), closeConnection(), patientLoad().
2. Connect patientLoad using AbstractPatientDatabaseLoader.cpp to load the patient.
3. Connect patientLoad using Adapter.cpp
4. The PatientManagementsystem call composite file and load, initial, close function.

Git commits:

1. I first added Composite class in commit 612763.
2. I fixed something in commit 8a0e00b.

```

commit 61276369649ea2435b33e957e12233e9b51d7fbc
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date: Mon Jun 5 20:02:25 2023 +0900
  
```

F1~F4- Draft of Design Pattern.

```

commit 8a0e00b43548b3882680611879ed2ef843a76662
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date: Thu Jun 8 20:13:30 2023 +0900
  
```

F1 - Adapter and Composite design pattern

Functional requirement FR3: calculate the alert level for all diseases a patient has

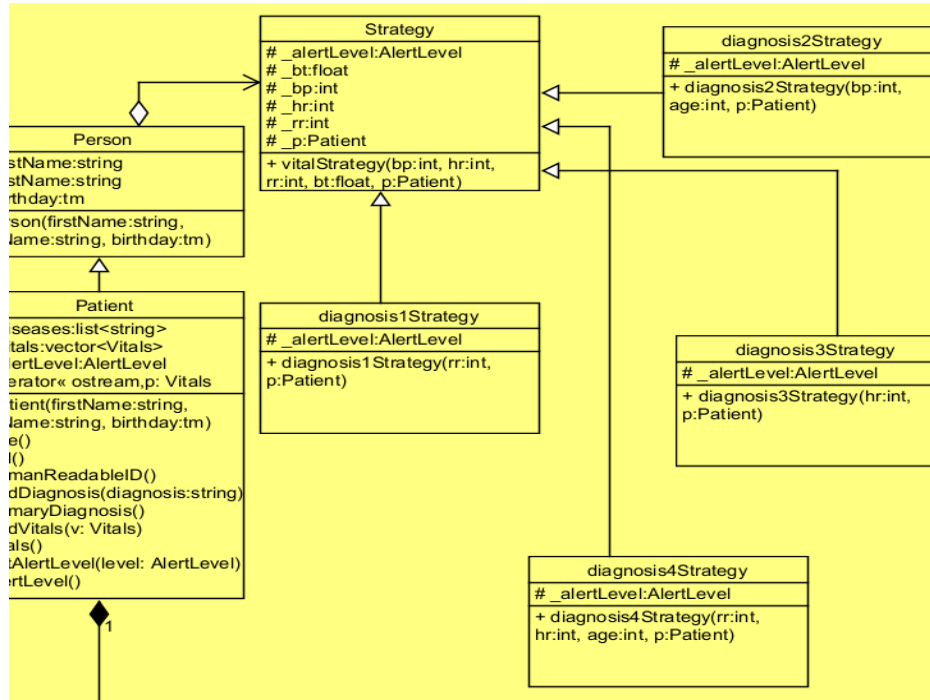


Figure4: Strategy Pattern

Design Patterns: Strategy Pattern

A patient has more than one disease. The system has to calculate a patient's highest alert level dependent on all their disease. Each patient use all algorithms and alert level must be the highest level calculated using Vitals or not be calculated for historical patient data loaded from a database or file using strategy pattern. Consists of Strategy pattern are 'Strategy', 'Diagnosis1Strategy', 'Diagnosis2Strategy', 'Diagnosis3Strategy', 'Diagnosis4Strategy'.

How it works:

1. Include other whole strategy classes such as 'Diagnosis1Strategy', 'Diagnosis2Strategy', 'Diagnosis3Strategy', 'Diagnosis4Strategy'.
2. Check the patients have what diseases from stored data
3. AlertLevel - A patient defaults Green alert level unless their vitals raise their level.

Git commits:

1. I first added Strategy class in commit 612763.
2. I created strategy classes in commit in 362ef7.
3. Other commits include: e58965, f34ec39, 1692f2, b6187c, 873ef2.

```
commit 61276369649ea2435b33e957e12233e9b51d7fbc
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date: Mon Jun 5 20:02:25 2023 +0900
```

F1~F4- Draft of Design Pattern.

```
commit 362ef7a8fe2f7cf9a72f5d0f68ecef41a69a9b59
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date:   Fri Jun 9 13:40:12 2023 +0900
```

FR2 - Strategy.cpp and Strategy.h in order to calculate the patient alert level.

```
commit e589656c8c54d49c29cb394c36377b42dc7b746b
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date:   Fri Jun 9 14:21:47 2023 +0900
```

FR3 - Create Diagnosis1Strategy.cpp and header file like Bonus Eruptus to calculate the alert level for all diseases a patient has.

```
commit f34ec396c065da279baa6364d6e713d969b88258
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date:   Fri Jun 9 14:22:24 2023 +0900
```

FR3 - Create Diagnosis2Strategy.cpp and header file like Amoria Phlebitis to calculate the alert level for all diseases a patient has.

```
commit 1692f2d0254b0c93338fc2f794ad58935b6e7911
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date:   Fri Jun 9 14:22:45 2023 +0900
```

FR3 - Create Diagnosis3Strategy.cpp and header file like Mad Zombie Disease to calculate the alert level for all diseases a patient has.

```
commit b6187c657903072c4dd9bd78ade5a7104694b553
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date:   Fri Jun 9 14:35:33 2023 +0900
```

FR3 - Create Diagnosis4Strategy.cpp and header file like Three Stooges Syndrome to calculate the alert level for all diseases a patient has.

```
commit 873ef27d7b12f5b9c617c0bb18ad3eb32168ce85
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date:   Fri Jun 9 15:49:18 2023 +0900
```

FR2, FR3 - Fix something for Diagnosis4Strategy.cpp and Strategy.cpp

Functional requirement FR4: alert the hospitals and GPs

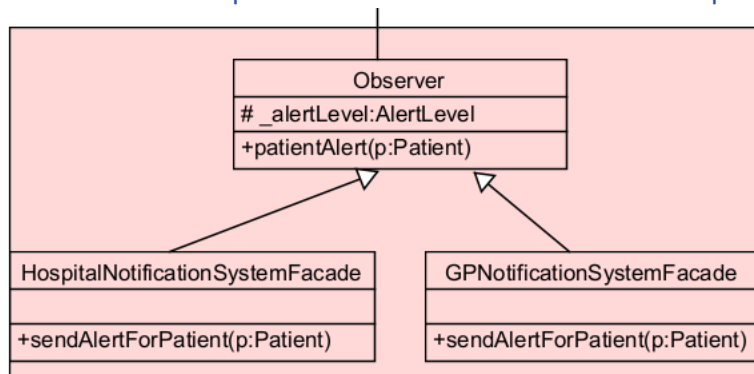


Figure5: Observer Pattern

Design Patterns: Observer Pattern

System should check stakeholders when a patient's alert level reaches Red. Interfaces are provided that is able to alert hospitals and GPs of concerning patients. Connect these facades to changes in patient alert levels. This notification has to happen immediately when a patient's alert level changes using Observer design pattern.

How it works:

1. Using patientAlert() in observer class, this notification has to happen immediately when a patient's alert level changes using Observer design pattern.
2. System should check stakeholders when a patient's alert level reaches Red. System should display a message to the GP and hospital using GPNotificationSystemFacade class and HospitalNotificationSystemFacade class.

Git commits:

1. I first added Observer class in commit 612763.
2. I also fixed something in commit c39e7a.

```
commit 61276369649ea2435b33e957e12233e9b51d7fbc
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date: Mon Jun 5 20:02:25 2023 +0900
```

F1~F4- Draft of Design Pattern.

```
commit c39e7a2098eef51d596ca176616edc767f41b501
Author: Gitae Bae <baegy002@mymail.unisa.edu.au>
Date: Fri Jun 9 15:02:59 2023 +0900
```

FR4 - Create Observer.cpp and headerfile to alert the hospitals and GPs.